

Math 170S - Homework 4 - due February 6

Problem 1. Your friend rolls a fair die, defines θ to be the number on the die, but does not tell you what is θ . But now she can give you samples X_i 's according to the distribution

$$f(x|\theta) = \theta \text{ for } 0 \leq x \leq 1/\theta$$

and 0 otherwise.

- a) If you get a sample $X = 0.3$, what would you guess θ is? Give posterior distribution and MAP estimate for θ .
- b) You ask for more independent samples to refine your estimate, and you get $X_1 = 0.3$, $X_2 = 0.4$, $X_3 = 0.1$. What is your estimate for θ now?

Hint: your estimates in a) and b) should be different.

Recall from class, that MAP estimate is defined as

$$\hat{\theta} = \operatorname{argmax}_{\theta} f_{\Theta|X}(\theta|x).$$

Problem 2. Assume that the observation X is normal distributed with unknown mean θ and known variance 9. Assume that the prior f_{θ} is normal with mean 5 and variance 4. You observe $X = 1$. Calculate the posterior distribution of θ . What is the MAP and conditional expectation estimates of θ ?

Recall from class, that the conditional expectation estimate is defined as

$$\hat{\theta} = \mathbb{E}f_{\Theta|X}(\theta|x).$$

Problem 3. You observe 21 data points measuring the temperature at UCLA on a daily basis. The results yielded a sample mean of 72.3 and a sample variance of 5.6 degrees Fahrenheit.

- a) Use Student's distribution to calculate a 90% confidence interval.
- b) Calculate the approximate confidence of the interval [70.5, 73].
- c) Calculate a one sided 99 % confidence interval that gives an upper bound on the mean temperature.
- d) Given you know that the real variance is 5 degrees, what would be the answers to part (a), (b) and (c) using the normal distribution?

Problem 4. 7.1-10

- d) Find 80% two-sided confidence interval for μ .

Problem 5. Let $\bar{X}$ be the mean of a random sample of size n from $N(\mu, 9)$. Find n so that $\mathbb{P}(\bar{X} - 1 < \mu < \bar{X} + 1) = 0.9$.

Problem 6. 7.2-10